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Short communication

Study on the keyhole oscillation mechanism of laser welding based on electro-mechano-acoustical analogy theory

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Highlights

- A novel keyhole oscillation model was constructed by using the “electro-mechano-acoustical” analogy method.
- Resonant frequency and magnification amplification factor of the keyhole oscillation were derived from the proposed model.
- The extracted sensing features provided strong validation for the derived formulas.
- The derived formulas in turn analyzed the keyhole oscillation mechanism reflected by the multi-sensing data.

4. Theoretical analysis and explanation

5. Conclusions

CRediT authorship contribution statement

Declaration of Competing Interest

Acknowledgement

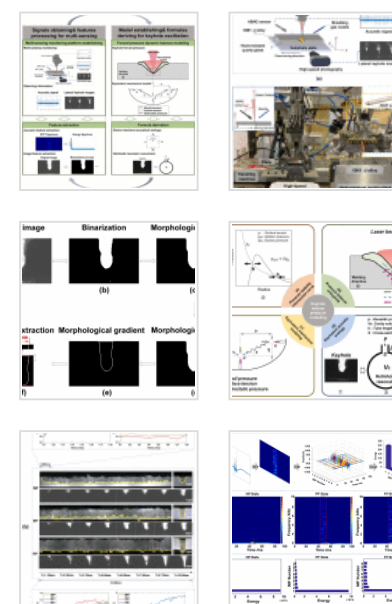
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Abstract

The acoustic emission phenomena generated by the laser welding process are closely related to the keyhole's forced oscillation and the dynamic pressure balance at the keyhole bottom. In this study, we constructed the mechanical equation of keyhole bottom pressure balance and used the “electro-mechano-acoustical” analogy theory to convert it into an acoustic equation, the acoustic resonant frequency as well as the magnification amplification factor of the keyhole oscillation in different penetration states were derived. A monitoring platform combining acoustic sensor and high-speed photography was established to obtain acoustic emission signals and lateral keyhole images. The acoustic and image features extracted from the monitoring data validated the derived formulas. Additionally, the acoustic phenomena of different frequency-energy components were explained according to the derived formulas in turn. The results demonstrated that the derived theory is compatible with experimental comparisons, showcasing its great interpretability and applicability. The introduced analogy theory effectively establishes a bridge between keyhole forced pressure and acoustic emission vibration, aiding in the understanding of keyhole oscillation characteristics. A novel model of keyhole oscillation was proposed and verified using process monitoring data in this study, which provides a new approach for the investigation of keyhole dynamic.

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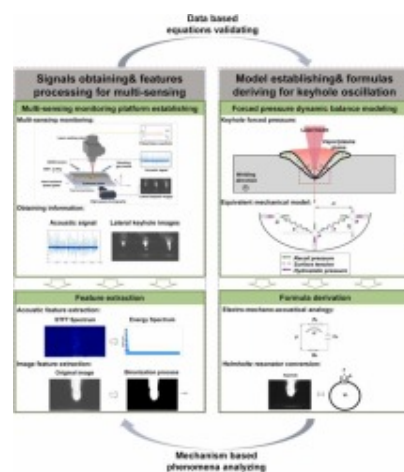


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Graphical Abstract



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Keywords

Laser welding; Keyhole oscillation; Electro-mechano-acoustical analogy; Acoustic characterization; Process monitoring

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1. Introduction

Laser welding has been widely used in the manufacturing fields of various lightweight alloys joining [1], [2] and dissimilar metals joining [3], [4] due to its unique advantages in high energy density and high efficiency. However, the intense molten pool fluctuation [5] and keyhole oscillation [6] of laser welding directly affect the formation quality of the

weld. Therefore, the study of the dynamic behavior of the keyhole and molten pool inside laser welding is of direct guidance for improving the weld quality.

In particular, the study of the keyhole dynamics behavior of laser welding provides the critical key to reveal the mechanism of keyhole oscillation. The unique keyhole oscillation behavior formed by the laser processing has been investigated by many researchers. Klein et al. in researches 7 and [8] modeled the radial, axial, and azimuthal oscillation of the keyhole, and proposed a model for the keyhole oscillations in response to power fluctuations of laser beams. However, Klein's model assumed that the keyhole is cylindrical, which had some limitations due to equipment and conditions. Volpp and Vollertsen [9] investigated the relationship between the oscillation frequency of the keyhole and the intensity distribution of the beam, and constructed the static and dynamic keyhole profile models. In order to investigate the interaction between the keyhole oscillation and metal vapor, Li et al. [10] constructed a glass-workpiece structure with a high-speed camera to observe the keyhole forms, and found that the folds on the front wall of the keyhole and the vapor-generated waves on the back wall of the keyhole were closely related to the vapor fluctuations inside the keyhole and the spattering at the openings of the outside keyhole. Besides the study of the internal dynamics of the keyhole, the vapor plume outside the keyhole is also closely related to the keyhole oscillation. Huang et al. [11] investigated the correlation between the dynamic behaviors of the vapor plume generated by the laser welding of 5083 aluminum alloy and the oscillations of the keyhole, and the correlation between the velocity oscillations of the plume field and the keyhole size is discussed. Wang et al. [12] concluded that the plasma/metal vapor induced by high power generates periodic oscillations above the molten pool, and the vibrations of the keyholes are closely related to the oscillations. Zhang et al. [13] investigated the frequency domain characteristics of keyhole area and centroid high plasma plume variation, and the correlation between penetration states and oscillations have been discussed. Zhang et al. [6] concluded that the width variation of the keyhole reaches the steady state earlier than the depth, while the high recoil pressure generated by the front wall bulge is the main reason for the oscillation of the

back wall. Although the phenomenon of keyhole oscillations has been investigated in many studies, the mechanism of this phenomenon is not well summarized.

The Air-borne Acoustic Emission (ABAE) phenomenon of metal vapor and plasma driven by the oscillation mechanism at the keyhole bottom is closely related to the dynamic behavior of the keyhole, and the monitoring of this acoustic phenomenon can help to reveal the keyhole defect forming mechanism. Zhao et al. [14] suggested that the critical destabilization at the bottom of the keyhole releases acoustic waves that can drive air bubbles away from the keyhole and form porosity defects. Schmidt et al. [15] investigated the acoustic wave correlation with spattering in laser welding. Gourishetti et al. [16] investigated the correlation between the acoustic emission phenomenon and the formation of gaps in laser welding. M. Yusof et al. [17], [18] used acoustic monitoring combined with machine learning to predict the depth of penetration in laser welding and classify the penetration states. In the study of these acoustic signals, the acoustic emission phenomena from keyholes are categorized into Structural-borne Acoustic Emission (SBAE) and ABAE, and in the review by M. Yusof et al. [19], the main sources of the ABAE signals from laser welding are considered as related to the oscillations of the plasma plume, the molten pool, and the keyholes. Among them, the unique energy and spectrum information embedded in the laser welding ABAE signals are relevant to the penetration states of the weld. Kovacevic [20] investigated the difference in acoustic spectrum between Full Penetration (FP) and Partial Penetration (PP) states in the low-frequency band (500–1500 Hz), and it was found that the power spectrum density of the FP state is higher than that of the PP state. Duley and Mao [21] investigated the strong correlation between the acoustic emission spectral range in the 20–20,000 Hz band and the penetration of the welds in CO₂ laser welding. Bastuck et al. investigated the variation of the SBAE [22] and ABAE [23] signals in higher frequency bands between the different penetration states in the laser welding of zinc-coated steels, and the study showed that the high-frequency AE signals were not detected during the conduction welding stage and before reaching the deep penetration welding stage. However, little research has been

seen on the use of acoustic sensing to monitor such oscillations and investigate their acoustic characteristics.

In summary, the keyhole oscillations are closely related to the acoustic emission phenomenon as well as the penetration states, but the connection between the principles of keyhole oscillation and the acoustic characteristics have not been clearly summarized and studied so far. Therefore, it is necessary to review the mechanism of keyhole oscillation in laser welding, and to discover the most essential correlation between keyhole oscillation and acoustic wave signals. Furthermore, how to establish a bridge between the monitoring technique and the mechanism study is the key to solve the problem.

In recent years, the “electro-mechano-acoustical” analogy method has been applied to laser welding because of its easy analogy conversion between electrical, mechanical, and acoustic systems. Shan He et al. [24] firstly made an analogy between the laser welding keyhole in deep penetration welding, established an electrical roadmap by analogizing the acoustic frequency law at the opening of the keyhole. However, the laser welding process is so complicated that the formation of the keyhole is under the action of various pressures, and the dynamic balance of the various pressures will cause the changes of the keyhole profiles at different states, which makes it insufficient to universally generalize the oscillatory mechanism and frequency characteristics of the keyhole just by using the electrical roadmap analogy for the keyhole. Therefore, it is necessary to establish a suitable dynamic pressure balance model for keyhole oscillation, accompanied by accurate monitoring methods to study the morphology and oscillations of keyholes in different states.

In this study, we proposed a keyhole bottom oscillation model based on the dynamic pressure balance principle, the induced “electro-mechano-acoustical” analogy theory was used to convert mechano-acoustical circuit and derived the equations of resonant frequency and magnification amplification factor of keyhole oscillation, which can fit the time-frequency characteristics of acoustic signal performance well, and the proposed

theory has logical self-consistency in accordance with the physical laws of laser welding. In the proposed method, firstly, a process monitoring platform was built to collect ABAE signals and lateral keyhole profiles respectively; secondly, an oscillation model based on “electro-mechano-acoustical” analogy theory was constructed based on the pressures balance at the bottom of the keyhole; and at last, the theoretical validity of the theory is verified by the monitoring features, and the phenomena of the monitoring were analyzed by the mechanism-based derivations.

This study provided a novel theoretical model for the keyhole oscillation mechanism of the laser welding process, and provided frequency characteristics and magnification amplification factor references for the diagnosis of the penetration states, which can help to further reveal the driving mechanism of the keyhole oscillation and improve the monitoring accuracy.

2. Experimental setup and theoretical foundation

To ensure the capture of acoustic waves generated by keyhole oscillation and the corresponding keyhole profiles during the laser welding process, a process monitoring platform combining with ABAE sensing and lateral high-speed photography was constructed. Additionally, the impedance analogy equations of mechano-acoustical analogy theory and the reference equations for different cavities are provided.

2.1. Experimental equipment and platform construction

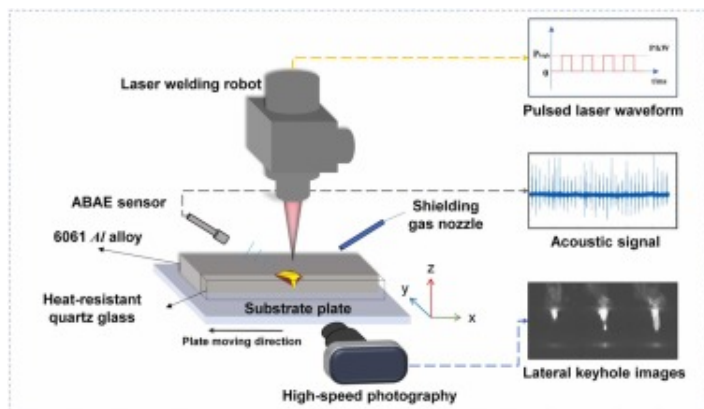
The 6061 aluminum alloy was used as the weld metal and each size is 100 mm×50 mm×3 mm. The laser equipment used in this study was a TruPluse 556 fiber pulsed laser produced by TRUMPF, with the laser wavelength at 1064 nm and a maximum power of 10 kW. The laser head (Trumpf BEO D70) was connected to the laser with a deflection angle of 5° to avoid laser reflection. The focus of the laser head was 223 mm from the metal surface, pure argon was used as the shielding gas, a rectangular pulse waveform was selected, and the beam diameter at the focus position was 0.6 mm. The

laser head was fixed, and the workpiece was put on the substrate plate driven by a traveling mechanism, with the speed of the traveling mechanism being 20 mm/s .

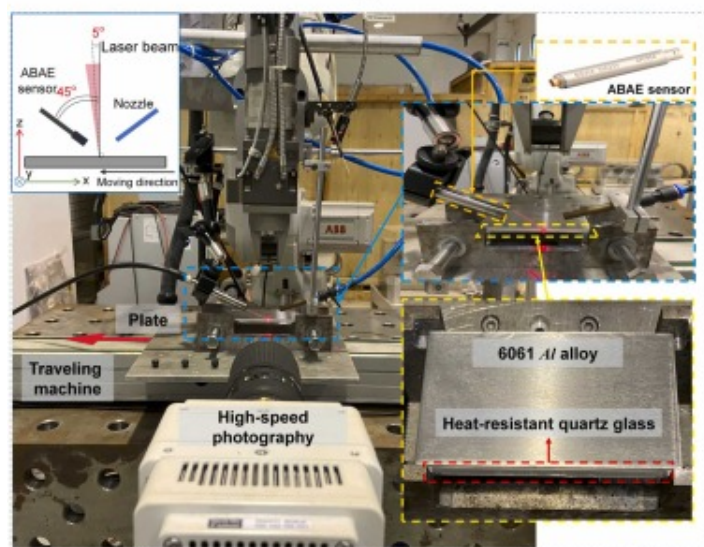
The glass for “glass-workpiece” structure was used with heat-resistance quartz glass of JGS2. Lateral high-speed photography equipment was used with a model Phantom VEO series 710L high-speed camera from Vision Research, an American company, fitted with an 810 nm filter designed to reduce the splatter and glare interference. The sampling frequency was set to $10,000\text{ fps}$ and the exposure time was $1\text{ }\mu\text{s}$. The original keyhole image was $512 \times 384\text{ pixels}$ with a resolution of 0.022 mm/pixel .

The ABAE acquisition equipment used in the experiment was the MA231 Preamplifier microphone transducer produced by BSWA TECH Company, and the acquisition card used the EM9118B series products produced by ZTIC Science and Technology Company, and the sampling rate S_p was set to 20 kHz .

The distance between the high-speed photography and the “glass-workpiece” structure is 400 mm , the distance between the airborne acoustic emission sensor and the workpiece is 100 mm , and the angle is fixed at 45° . When the pulsed laser works, the compressed gas and argon gas are activated at the same time, the high-speed photography starts the capture program, and the acquisition card starts to capture the welding acoustic wave. The schematic diagram of the experiment is shown in [Fig. 1](#). And the details of welding parameters used in the experiments and the corresponding penetration states are given in [Table 1](#).



(a)



(b)

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Fig. 1. View of laser welding monitoring platform. (a) Schematic diagram of experimental equipment; (b) Actual image of experimental equipment.

Table 1. Details of welding parameters and relevant penetration states. (NP, PP, FP represent Non-Penetration, Partial-Penetration and Full-Penetration, respectively).

Case	Laser Power	Pulse Width	Pulse Frequency	Welding Speed	Defocusing Distance	Penetration States
1	5 kW	10 ms	10 Hz	20 mm/s	0 mm	FP
2	4 kW	10 ms	10 Hz	20 mm/s	0 mm	PP
3	3 kW	10 ms	10 Hz	20 mm/s	0 mm	NP

2.2. Theoretical foundations of electro-mechano-acoustical analogy

The concept of “electro-mechano-acoustical” analogy is mainly used in electroacoustics, because the electroacoustic device is a coupled system of electrical, mechanical, and acoustical engineering. And the equivalent circuit diagrams derived from the electro-mechano-acoustical analogy can couple the three components together, allowing a more direct analysis of the induced response relationship.

The review article by Bertuccio [25] investigated the development history and derivation for the theory of the “electro-mechano-acoustical” analogy. It is argued that in the impedance analogy the laws of inductors and capacitors can be related to the mechanical mass, acoustic mass, spring laws, and acoustic compliance, respectively.

Over the years, many studies are still using “electro-mechano-acoustical” analogy theory to solve problems in the field of acoustics. Sun et al. [26] used the “electro-mechano-acoustical” analogy theory to investigate that the acoustic coupling between the linear alternators and the traveling-wave thermoacoustic engine is crucial for the performance of the system. Bai et al. [27] used the theory to investigate the acoustic properties such as vibration and acoustic radiation characteristics in the design of a flat panel loudspeaker.

According to the summary of the research [25], the impedance analogy formulas for electrical, mechanical, and acoustical can be converted to each other by analogy. For the purpose of calculation, the acoustical equations for the Helmholtz resonant cavity and open-closed tube in the “electro-mechano-acoustical” analogy theory [28] are given as

Table 2. where M_m is the force mass of the mechanical system, which in the acoustic system is equivalent to the mass of the medium, R_m is the force resistance of the mechanical system, c_0 is the speed of sound in the medium, ρ_0 is the density of the medium, η is the shear viscosity coefficient of the medium, V is the volume of the cavity, S is the cross-sectional area of the short tube, d is the diameter of the tube or the hole, and l is the length of the tube (in the case of the Helmholtz resonator with the inserted short tube $l=0.61d$, and without the short tube $l = 0.85d$, and when there is only a hole $l=0.85d$).

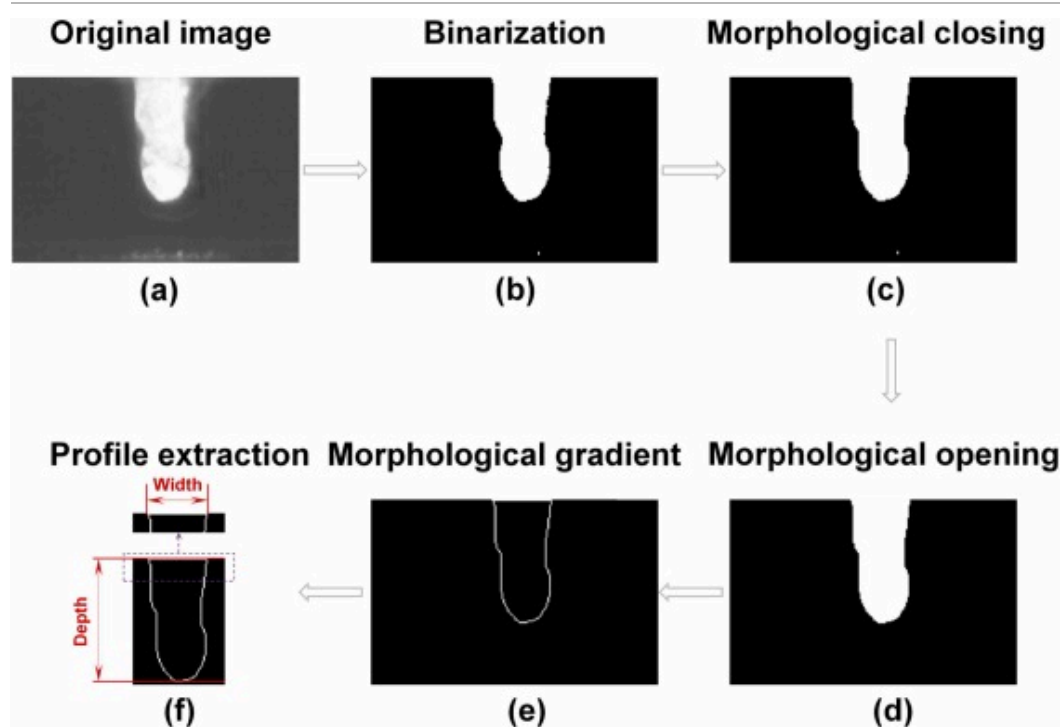
Table 2. Acoustic element formulas in different profiles.

Electrical element (quantity)	Cavity	Open-closed Tube
Acoustic Compliance (C_a)	$C_a = \frac{V}{\rho_0 c_0^2}$	$C_a = \frac{\pi d^2 l}{4 \rho_0 c_0^2}$
Acoustic Mass (M_a)	$M_a = \frac{M_m}{S^2}$ $= \frac{\rho_0 l}{S}$	$M_a = \frac{4 \rho_0 (1+0.85d)}{\pi d^2}$
Acoustic Resistance (R_a)	$R_a = \frac{R_m}{S^2}$	$R_a = \frac{4 \rho_0 \sqrt{2 \eta \omega}}{\pi d^2} \left(\frac{2l}{d} + 2 \right)$

2.3. Foundations of image processing algorithms

In order to measure the depth and width of the keyhole images captured in high-speed photography, an image processing algorithm is introduced. The steps of the image processing algorithm to extract the keyhole features are as follows:

- The original image ([Fig. 2\(a\)](#)) is first binarized using the binarization image algorithm ([Fig. 2\(b\)](#));



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Fig. 2. An overall image processing pipeline of extracting the keyhole depth and width.

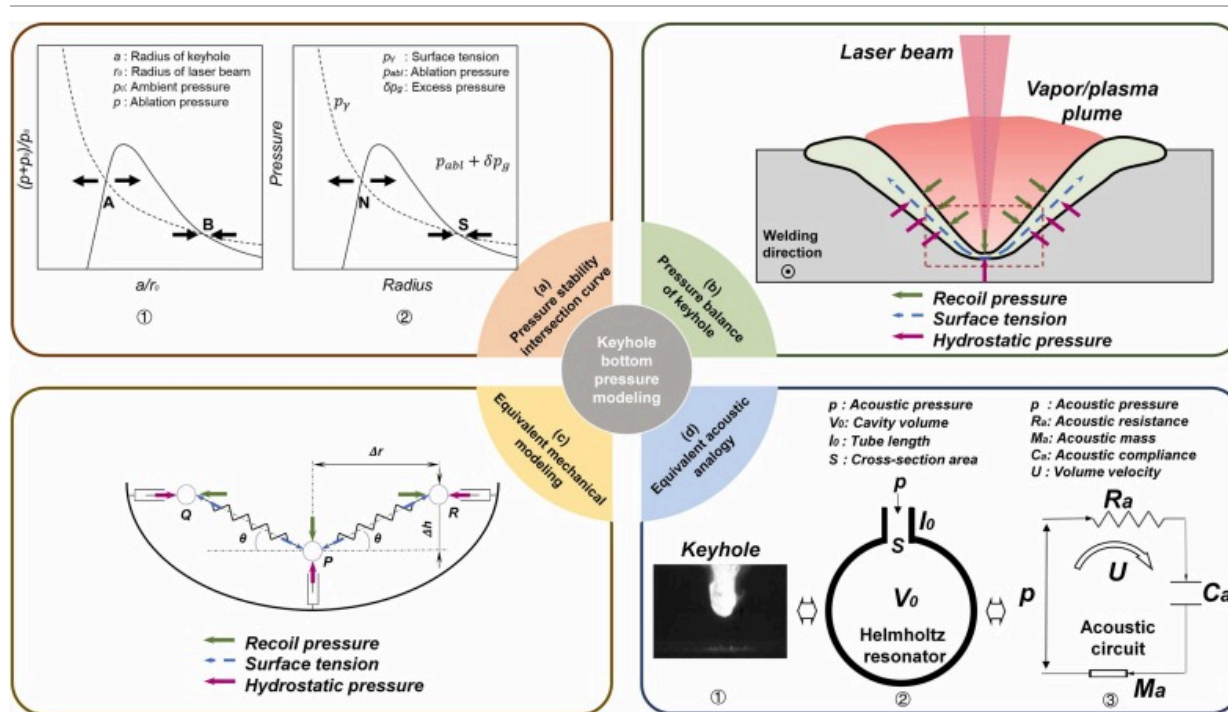
- b) Apply the morphological closing algorithm to fill the binarized image and to exclude interference from pores and splatters (Fig. 2(c));
- c) Apply the morphological opening algorithm to remove the image noise (Fig. 2(d)), and the edge contour of the keyhole wall is obtained by the morphological gradient processing algorithm (Fig. 2(e));
- d) The depth Δh and the width Δd of the keyhole can be obtained by taking a localized cropping of the keyhole edge and extracting the depth and width using the external rectangle algorithm on the keyhole profile (Fig. 2(f)).

3. Derivation of equations for keyhole oscillation

By further summarizing the characteristics of pressure balance at the keyhole bottom, the oscillation mechanism of the keyhole can be better understood.

3.1. Derivation of resonant frequency

Variations in the depth and width of the keyhole were demonstrated by the keyhole oscillation. The studies of [29] and [30] were the first to summarize a law that demonstrating a functional relationship between the ratios of pressures-such as surface tension, recoil pressure and atmospheric pressure-and the ratios of the keyhole opening radius as well as the laser beam radius, as illustrated in the sub-image① of Fig. 3(a). Subsequently, the study [7] summarized a more concise plot of the pressure versus radius variation function based on this law, as shown in the sub-image② of Fig. 3(a). The *A* and *N* points are unstable balance points and the *B* and *S* points are stable balance points. Deviation of the keyhole radius from any value other than its balance point induces a difference in recoil pressure and surface tension, and the direction of this difference near the points *B* and *S* runs in the opposite direction of the disturbance, enabling the keyhole with oscillations.



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Fig. 3. An overall of keyhole bottom pressure modeling. (a) Pressure stability intersection curve(sub-image①: Ablation pressure (—) and surface tension (---) for the stability of the intersection points; sub-image②: Ablation pressure together with excess pressure (—) and surface tension (---) for the plotted against radius); (b) Pressure balance of keyhole; (c) Equivalent mechanical modeling; (d) Equivalent acoustic analogy.

It is generally accepted that the principle of reflection, transmission, and absorption of the laser beam on the surface of metals is the result of the interaction of the electromagnetic field of the light wave with the free and bound electrons of the metal. And when the laser irradiates the metal material surface, due to the small energy of the photons, it usually acts only on the free electrons in the metal. Under the action of the laser's electromagnetic field, the electrons present an oscillatory motion, which is the

essential reason why the laser heats the material to form keyholes and the keyholes generate oscillations. It's not the unique opinion, for example, Gong et al. concluded in their book [31] that the absorption for material of laser beam is satisfied: "The regular motion of damped oscillation of the charged particles in the atom is described by the classical mechanics; that is to say, the charged particles are regarded as the oscillators following the classical mechanics. The oscillators vibrate as bound close to a certain equilibrium position by the elastic restoring force directly proportional to the displacement. In case of deviating from the balanced position, the oscillators will be acted by a restoring force. In terms of metal, oscillators in the metallic material are mainly free electrons. Absorption in the metal is mainly completed by the vibration of free electrons"

By analyzing the reason for the pressure on the laser welding keyhole, as shown in Fig. 3(b), the pressure on the bottom of the keyhole is generated by the dynamic balance among the recoil pressure of the metal vapor, the hydrostatic pressure, and the surface tension, respectively, which together result in the changes of the keyhole's depth and width. Based on the above conclusions, in this study, the smallest mass point at the keyhole bottom that can be subjected to the recoil pressure, the hydrostatic metal pressure and the surface tension during the deep penetration of laser welding is considered as an oscillator.

When the deep penetration welding occurs, the oscillator is vibrated by the recoil pressure, which can be regarded as an elastic force to make it vibrate, and when the oscillator deviates from the balance position, the hydrostatic pressure of the molten metal and the surface tension play a joint role in blocking and restricting the vibration of the oscillator, which can be regarded as a damping force (the laser welding speed in this study is lower, and the effect of the hydrodynamic pressure is ignored).

Due to the different amount of the metal vapors generated from the Knudsen layer, which leads to different recoil pressures, we assume that the recoil pressure at point P of the keyhole wall is F_r . To simplify the calculations, we set the recoil pressures at points Q and

R of the keyhole wall to be equal and the value of μF_r , μ is a scale factor, which is related to the angle θ as shown in Fig. 3(c).

Furthermore, the pressure on the keyhole is simplified as in Fig. 3(c), with point P being the mass oscillator M_{m1} at the bottom of the keyhole, and Q and R being the mass oscillators M_{m2} at the opening of the keyhole, respectively. F_r is the vapor recoil pressure, the value is related to the metal vapor vaporized by the laser beam. Assume that the surface tension F_t on the keyhole wall resisting the action of F_r is equivalent to be a spring force, that is:

$$F_t = R_m v_t \quad (1)$$

where R_m is the compliance of the mass point, v_t is the velocity of motion in the direction of the keyhole wall, due to the properties of the equivalent spring force, the velocity along the spring direction is the same at the ends of P and Q . According to the trigonometric function of the θ angle, the value of the velocity v_1 in the vertical direction at the point P is:

$$v_1 = v_t \sin \theta \quad (2)$$

The magnitude of the velocity v_2 in the horizontal direction at point Q is:

$$v_2 = v_t \cos \theta \quad (3)$$

Assume that the hydrostatic pressure F_s resisting the expansion of the mass point at the bottom of the keyhole is equivalent to a resistance force, that is:

$$F_s = \frac{1}{C_m} \int v_s dt \quad (4)$$

where $1/C_m$ is the damping coefficient at the mass point, and the magnitude of v_s at point P is equal to the velocity of motion v_1 in the vertical direction at the mass point of P .

Similarly, the magnitude of v_s at point Q is equal to the motion velocity v_2 in the horizontal direction at point Q .

Assume that Δr is the radial radius of the keyhole, Δh is the axial depth of the keyhole, and θ is the angle between the wall of the keyhole and the horizontal direction at the beginning stage, according to the geometrical equation, the following is obtained as:

$$\theta = \arctan\left(\frac{\Delta h}{\Delta r}\right) \quad (5)$$

During the beginning stage of laser welding at the Δt time, since the pressure at the bottom of the keyhole is in dynamic balance, the balance equations at the two mass points are listed by taking point P and point Q as the objects of study, respectively.

List the balance equation of the vertical direction at point P . Due to the θ angle, the sum of the vertical component forces of surface tension at point P is:

$$2F_t \sin \theta \quad (6)$$

List the balance equation in the horizontal direction at point Q . The horizontal component of the surface tension at point Q is:

$$F_t \cos \theta \quad (7)$$

Define the velocity of the mass in the vertical direction as v_1 and the velocity of the mass in the horizontal direction as v_2 , respectively. And list the equations as follows.

In the vertical direction:

$$F_r - 2F_t \sin \theta - F_s = M_{m1} \frac{dv_1}{dt} \quad (8)$$

Solving the equation yields the formula as follows:

$$F_r - 2R_m v_1 - \frac{1}{C_m} \int v_1 dt = M_{m1} \frac{dv_1}{dt} \quad (9)$$

In the horizontal direction:

$$\mu F_r - F_t \cos \theta - F_s = M_{m2} \frac{dv_2}{dt} \quad (10)$$

Solving the equation yields the formula as follows:

$$\mu F_r - R_m v_2 - \frac{1}{C_m} \int v_2 dt = M_m \frac{dv_2}{dt} \quad (11)$$

Combining the balance equations for the vertical and horizontal directions and solving yields the formula as follows:

$$F_r = \frac{R_m}{1+\mu} (2v_1 + v_2) + \frac{1}{(1+\mu)C_m} \int (v_1 + v_2) dt + \frac{M_m}{1+\mu} \left(\frac{dv_1}{dt} + \frac{dv_2}{dt} \right) \quad (12)$$

By substituting the following formula:

$$F_r = \frac{R_m}{1+\mu} (2v_t \sin \theta + v_t \cos \theta) + \frac{1}{(1+\mu)C_m} \int (v_t \sin \theta + v_t \cos \theta) dt + \frac{M_m}{1+\mu} \left(\frac{dv_t \sin \theta}{dt} + \frac{dv_t \cos \theta}{dt} \right) \quad (13)$$

Solving the equation yields the formula as follows:

$$F_r = \frac{\sqrt{2}}{1+\mu} \sin \left(\theta + \frac{\pi}{4} \right) \left(\frac{\sqrt{10}}{2} R_m v_t + \frac{1}{C_m} \int v_t dt + M_m \frac{dv_t}{dt} \right) \quad (14)$$

Provide the equivalent of surface tension in the spring force, the spring undergoes simple harmonic vibration with a velocity of

$$v_t = v_0 e^{j\omega t} \quad (15)$$

where ω is angular frequency of the spring unit in the complex domain, and since the angle θ is an angle in the real domain, for ease of computation, defining that:

$$\omega = \alpha \omega_0 \quad (16)$$

where ω_0 is the equivalent real-domain angular frequency, and α is assumed as a fixed scaling factor.

Solving the equation yields the formula as follows:

$$F_r = \frac{\sqrt{2}}{1+\mu} \sin\left(\theta + \frac{\pi}{4}\right) \left(\frac{\sqrt{10}}{2} R_m + \frac{1}{j\omega C_m} + j\omega M_m \right) v_t \quad (17)$$

Since that,

$$Z_m = \frac{\sqrt{2}}{1+\mu} \sin\left(\theta + \frac{\pi}{4}\right) \left(\frac{\sqrt{10}}{2} R_m + \frac{1}{j\omega C_m} + j\omega M_m \right) \quad (18)$$

Z_m is the impedance, and solving the equation yields as follows:

$$Z_m = \frac{\sqrt{2}}{1+\mu} \sin\left(\theta + \frac{\pi}{4}\right) \sqrt{\left(\frac{\sqrt{10}}{2} R_m\right)^2 + \left(\omega M_m - \frac{1}{\omega C_m}\right)^2} e^{\left[j \arctan\left(\frac{\omega M_m - \frac{1}{\omega C_m}}{\frac{\sqrt{10} R_m}{2}}\right) \right]} \quad (19)$$

Applying a mechano-acoustical analogy conversion to the obtained impedance Z_m , the following is obtained as follows:

$$Z_a = \frac{\sqrt{5}}{1+\mu} \sin\left(\theta + \frac{\pi}{4}\right) R_a + \frac{\sqrt{2}}{1+\mu} \sin\left(\theta + \frac{\pi}{4}\right) M_a j\omega - \frac{\sqrt{2}}{1+\mu} \sin\left(\theta + \frac{\pi}{4}\right) \frac{j}{\omega C_a} \quad (20)$$

Defining that, the follow equations of

$$\frac{\sqrt{5}}{1+\mu} \sin\left(\theta + \frac{\pi}{4}\right) R_a = R_{a0} \quad (21)$$

$$\frac{\sqrt{2}}{1+\mu} \sin\left(\theta + \frac{\pi}{4}\right) M_a = M_{a0} \quad (22)$$

$$\frac{\sqrt{2}}{1+\mu} \sin\left(\theta + \frac{\pi}{4}\right) \frac{1}{C_a} = \frac{1}{C_{a0}} \quad (23)$$

are the equivalent acoustic resistance, equivalent acoustic mass and equivalent acoustic capacitance, respectively. Then the equivalent acoustic impedance becomes

$$Z_{a0} = R_{a0} + j\left(M_{a0}\omega - \frac{1}{\omega C_{a0}}\right) \quad (24)$$

It can be seen the acoustic circuit diagram equals to a Helmholtz resonator as shown in the sub-image② in Fig. 3(d), in general, the Helmholtz resonator is composed of a short

tube and a cavity, which can be stimulated by the external acoustic field to absorb and consume its energy, while the vibration inside the cavity can be radiated and transmitted through the short tube to strengthen the acoustic field outward. According to the electro-mechano-acoustical analogy conversion, the acoustic pressure expression can be obtained as follows:

$$p = U \left[R_{a_0} + j \left(M_{a_0} \omega - \frac{1}{\omega C_{a_0}} \right) \right] \quad (25)$$

where p is the acoustic pressure and U is the volume velocity, an acoustic circuit can be made as the sub-image③ in Fig. 3(d).

It follows from the same logic that when

$$\omega M_{a_0} = \frac{1}{\omega C_{a_0}} \quad (26)$$

the following formula is obtained as:

$$f_0 = \frac{1}{2\pi\alpha\sqrt{M_{a_0}C_{a_0}}} \quad (27)$$

where Z_{a0} is at its minimum and U is at its maximum, that is, the system resonates, or the acoustic resonance occurs. It can be obtained at this time that the equivalent resonant frequency f_0 analytical formula is:

$$f_0 = \frac{1}{2\pi\alpha\sqrt{\left[\frac{\sqrt{2}}{1+\mu}\sin\left(\theta+\frac{\pi}{4}\right)M_a \right] \left[\frac{C_a}{\frac{\sqrt{2}}{1+\mu}\sin\left(\theta+\frac{\pi}{4}\right)} \right]}} = \frac{1}{2\pi\alpha\sqrt{M_a C_a}} \quad (28)$$

When the keyhole profile appears as the resonator profile, according to Table 2, the keyhole is approximated to be a Helmholtz resonator without a short tube but only a hole, and the resonant frequency at this point is:

$$f_{a_0} = \frac{1}{2\pi\alpha\sqrt{M_a C_a}} = \frac{c_0}{2\pi\alpha} \sqrt{\frac{S}{Vl}} = \frac{c_0}{2\pi\alpha} \sqrt{\frac{6\Delta r}{1.7(\Delta h)^3}} \quad (29)$$

When the keyhole profile appears as the narrow short tube, according to Table 2, the keyhole is approximated as the open-closed tube, and the resonant frequency at this point is:

$$f_{a0} = \frac{1}{2\pi\alpha\sqrt{M_a C_a}} = \frac{c_0}{2\pi\alpha\sqrt{(1+0.85d)l}} = \frac{c_0}{2\pi\alpha\sqrt{(1+1.7\Delta r)\Delta h}} \quad (30)$$

3.2. Derivation of magnification amplification factor

When it is at the resonant frequency, there is only acoustic resistance R_{a0} , and if the acoustic pressure of the system is p , then the volume velocity is:

$$U = \frac{p}{R_{a0} + R_r} \quad (31)$$

where R_r is the radiation acoustic resistance of the opening. In order to explore the magnitude of the acoustic pressure at the keyhole opening, it is assumed that there is a moving piston at the keyhole opening, which is subjected to acoustic pressure, and according to the study [28], the radiation resistance of the piston is characterized as:

$$R_M = \pi a^2 \rho_0 c_0 \left[\frac{1}{2} (ka)^2 \right] \quad (32)$$

where a is the radius of the piston, i.e., the radius of the keyhole opening Δr , πa^2 is the cross-sectional area of the piston, i.e., the cross-sectional area of the keyhole opening $\pi \Delta r^2$, and $k=\omega/c_0$ is the propagation coefficient when there is no attenuation.

According to the conversion relationship between mechanical and acoustical resistance, the equivalent acoustic resistance of the piston is given as:

$$R_r = \frac{R_M}{S^2} = \frac{2\pi\rho_0\alpha^2 f_0^2}{c_0} \quad (33)$$

where f_0 is the value of the resonant frequency, because at this time the inside of the keyhole and the opening of the keyhole are connected, so the radiated acoustic resistance at the opening of the keyhole is equal to the value of the acoustic resistance of the

keyhole cavity, i.e., $R_a=R_r$, and the value of the volume velocity can be obtained as:

$$U = \frac{p}{2R_r} = \frac{c_0}{4\pi\rho_0\alpha^2 f_0^2} p \quad (34)$$

Defining that, the equation of

$$A_u = \frac{c_0}{4\pi\rho_0\alpha^2 f_0^2} \quad (35)$$

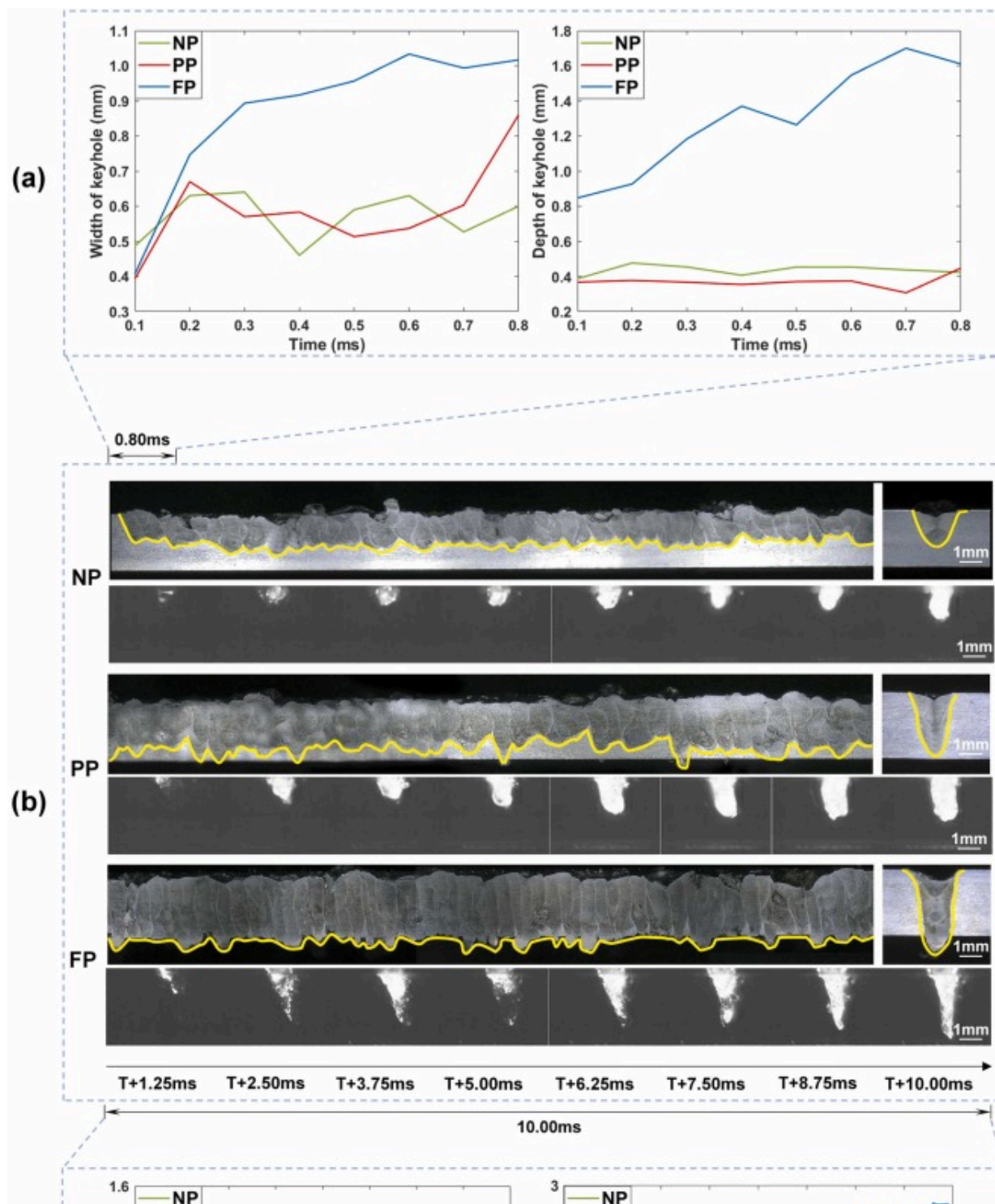
is the magnification amplification factor of the volume velocity U . The greater the volume is, the greater the velocity of the medium that flows through the keyhole opening, which means that the acoustic pressure at the keyhole opening is correspondingly greater. It can be deduced that the higher the acoustic pressure, the more intense the oscillations will be.

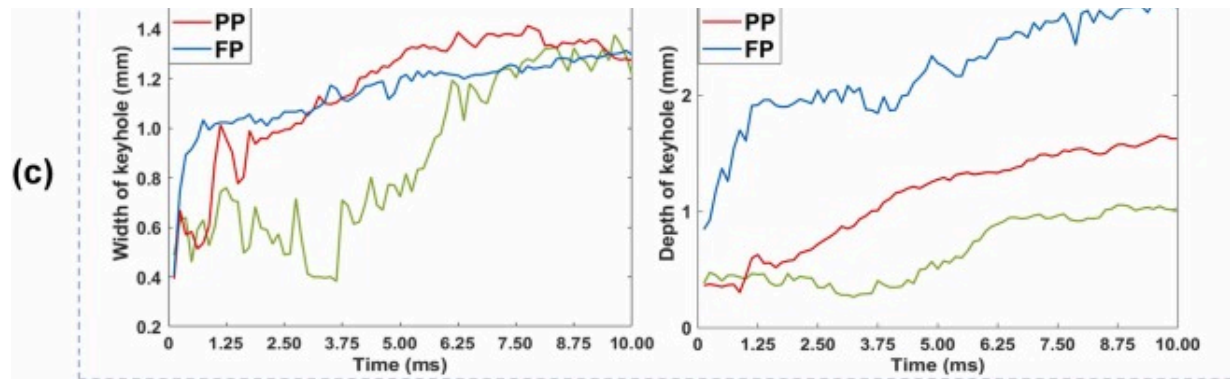
4. Theoretical analysis and explanation

To verify the consistency of the derived theory with the monitoring data, the image data were processed to extract the keyhole profile features, and the acoustic wave data were processed to extract the keyhole oscillation frequency features.

4.1. Validation on the image data

At the first 0.8 ms of the pulse cycle beginning, with each frame image every 0.1 ms, 50 pulse cycle of each penetration state were taken and the average value of the change in keyhole profiles was calculated, as shown in [Fig. 4\(a\)](#) and [Fig. 4\(b\)](#).





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Fig. 4. Variation curves of the keyhole width and depth in each state and the welding images of the keyhole and the welds. (a) Variation curves of keyhole width and depth from 0.00 to 0.80 ms in each state; (b) Keyhole images as well as cross-sectional welds images in each state; (c) Variation curves of keyhole width and depth from 0.00 ms to 10.00 ms in each state. (NP, PP, FP represent Non-Penetration, Partial-Penetration and Full-Penetration, respectively).

It can be seen in the beginning stage of deep penetration welding that, the penetration widths of the three states are all fluctuated about from 0 to 1 mm, and the penetration depths of the three states differ greatly. The keyhole profiles of NP and PP are like the concave state, which can be approximated as a Helmholtz cavity. But the penetration depths of the FP state varying drastically, which have already assumed the morphology of a narrow and thin tube at the beginning.

Based on the derivation of the pressure balance at the keyhole bottom, the physical meaning of the keyhole resonant frequency can be comprehended as the frequency of the oscillation at the keyhole bottom induced when the recoil pressure reaches the intrinsic frequency of the mass point at the keyhole bottom. When the keyhole is approximated to

a Helmholtz cavity, according to Eq. (29), since α is assumed as a fixed scaling factor, then

$\frac{c_0}{2\pi\alpha}$ is a constant value, while $\sqrt{\frac{6\Delta r}{1.7(\Delta h)^3}}$ varies with the area of the keyhole opening.

Analyzing the formula, it can be seen in the cavity stage that, the numerical value of the resonant frequency for the keyhole is determined by the width and depth, while in this stage, the change in the depth of the keyhole for the NP and PP states can be approximated to be regarded as a constant value, while the change in the width of the keyhole is more obvious, however, according to the Eq. (29), the power series of the depth is more decisive to the change in the resonant frequency. Therefore, it is deduced that the resonant frequency at the beginning of the NP and PP states is very low, and the resonance can be induced by the external input frequency at a very low level, causing the frequency response at this resonance frequency to be the largest, with the corresponding amplitude at such a high level, which represents that the cavity is able to amplify the acoustic energy.

Similarly, in the entire pulse cycle, with each frame image every 0.1 ms, 50 pulse cycle of each penetration state were taken and the average value of the change in keyhole profiles was calculated, as shown in Fig. 4(b) and Fig. 4(c).

It can be seen in the full stage of the deep penetration welding that, the variation of the maximum welding widths of the three states is very small, and the variation of the maximum welding depths of the three states is very large, as well as the maximum welding depth of the FP state is much larger than that of the other two states, with the depth of the keyhole is much larger, which is similar to a slender open-closed tube. In the stage where the keyhole is approximated to an open-closed tube, according to Eq. (30), α is a fixed scale factor, then $\frac{c_0}{2\pi\alpha}$ is a constant value, while $\frac{1}{\sqrt{\sqrt{(1+1.7\Delta r)\Delta h}}}$ that varies with the width and depth of the keyhole represents a variable.

Analyzing the derivation formula, it is seen that the numerical value of the resonant frequency for the keyhole is determined by the keyhole width and depth. In the stage of

deep penetration welding forming a narrow keyhole, the depth of the keyhole varies greatly and the width of the keyhole varies only slightly, so that the depth of the keyhole dominates the change of the resonant frequency, which causes the resonant frequency to become smaller, and the external input frequency can be induced to resonance when the input frequency is also very small, and at this time the frequency response becomes the largest, with a correspondingly higher acoustic energy.

4.2. Validation on acoustic data

To test the accuracy of the formulas derived above, signal processing and validation of the acoustic signals were performed. The method of frequency domain analysis was used, which is a more accurate and efficient means of analysis and provides more different insights compared to the time domain. In our study, Fast Fourier Transform (FFT) algorithm was applied to extract the frequency characteristics of the AE signal segment as shown in the following equation:

$$X(k) = \sum_{t=0}^{N-1} x(t) \left(\cos 2\pi k \frac{t}{N} - j \sin 2\pi k \frac{t}{N} \right) (k = 0, 1, 2, \dots, N-1) \quad (36)$$

where $X(k)$ is the computed data after FFT processing, $x(t)$ is the recorded acoustic signal, and N is the sampling point ($N=20\ k$). In order to directly reflect the distribution of free-frequency domain features in time, the acoustic signals belonging to different melting states are converted into two-dimensional spectrum maps using Short Time Fourier Transform (STFT). The corresponding “time-frequency-energy” information is combined with the keyhole behavior for the study. The acoustic signals are analyzed segment by segment to obtain the local spectrum of the acoustic signals by moving the window function $h(t)$ on the time axis. The STFT of the original signal $x(t)$ is defined as:

$$STFT(t, f) = \int_{-\infty}^{+\infty} x(t) h(t - \tau) e^{-j2\pi f\tau} d\tau \quad (37)$$

STFT(t, f) is the computed data after STFT processing, $x(t)$ is the recorded acoustic signal, $h(t-\tau)$ is the sliding window function, and τ is the sliding step, $\tau < t$. The three-dimensional

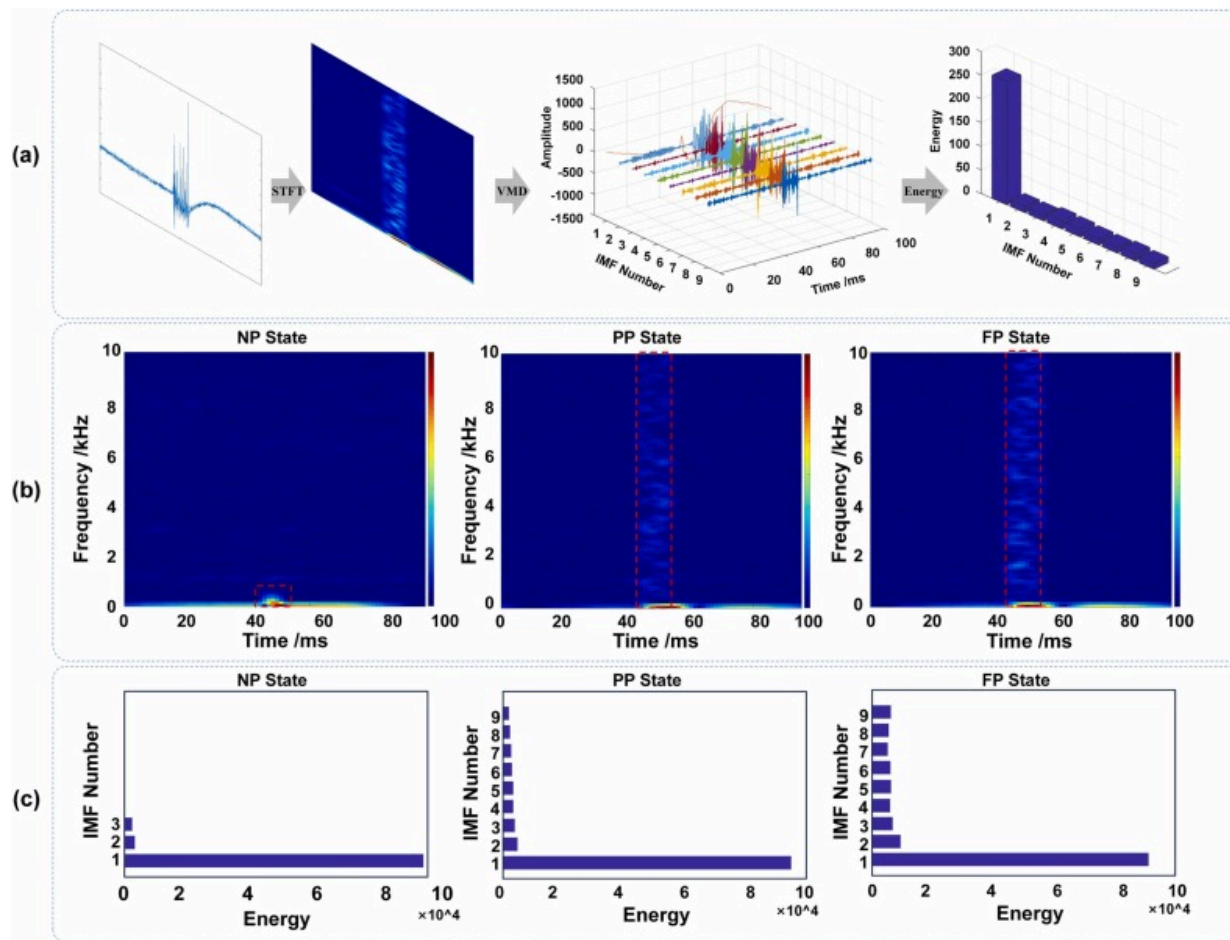
time-frequency-energy distribution of the original acoustic signal can be obtained from the graph.

The frequency decomposition of acoustic waves is performed by using the Variational Mode Decomposition (VMD) method, and the number of decomposition layers is determined using the instantaneous frequency averaging method according to our previous study [32]. The number of NP state decomposition layers of IMF is 3 and the number of PP and FP decomposition layers of IMF is 9.

In order to better represent the distribution of Intrinsic Mode Functions (IMF) frequency components, the time-domain energy of each IMF component is extracted using the same time-domain energy calculation method,

$$En(x) = \int_0^N |x(t)|^2 dt \quad (38)$$

where N is the maximum number of time series, the energy distribution of each IMF component is plotted to obtain the distribution of energy. The acoustic processing flow is shown in Fig. 5(a). Here, the STFT spectrum grams and Energy spectrum grams of each state are shown in Fig. 5(b) and Fig. 5(c).



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Fig. 5. The STFT-Energy distribution images. (a) The flowchart of acoustic processing; (b) The STFT spectrum distribution of single frame acoustic signal in each state; (c) The Energy distribution of entire period acoustic signal in each state. (NP, PP, FP represent Non-Penetration, Partial-Penetration and Full-Penetration, respectively).

As can be seen from Fig. 5(b) and Fig. 5(c), the low-frequency high-energy component (IMF 1) can be seen in each state, and the high-frequency low-energy components (IMF

6–9) almost merely exists in PP and FP states, no high-frequency components are found in NP state.

Firstly, the low-frequency high-energy component phenomenon is analyzed. According to the Eq. (29), the resonant frequencies are at a low level in both keyhole profiles, so it can be deduced that the low-frequency component is the frequency interval of the keyhole-induced resonance, according to frequency response principle, the maximum energy is generated at the frequency response, which explains the existence of high-energy component. It's not the unique phenomena only in this research, the research [13] found that the variation characteristics of centroid high plasma plume and the keyhole area are similar in the low-frequency component (0–4000 Hz), and the amplitude of frequency in the full penetration mode is higher than that in the partial penetration mode. The research [20] found that the power spectrum in FP state is greater than that in PP state in low-frequency. What's more, the research [33] showed that the deeper keyholes present higher spectrum intensity below 20 kHz.

Then, the high-frequency low-energy component phenomenon is also analyzed. Beyond the resonant frequency interval, the external input frequency to the bottom of the keyhole varies with the laser energy. Besides the beginning stage in pulsed interval, the high-energy laser continues to vaporize the metal and to generate high recoil pressure, the higher the recoil pressure is, the higher the equivalent pressure on the oscillator will be. And according to the balance condition of the pressure at the bottom of the keyhole, the hydrostatic pressure and the surface tension play a corresponding role in the recovery, and therefore the speed and frequency of oscillation will be greater. As a result, the output frequency of the oscillation system will be larger, and the width and depth of the keyhole will gradually increase, as well as the magnification amplification factor of the acoustic pressure will also increase accordingly, which means that the keyhole needs to continuously absorb energy to expand the hole. However, the frequency response of the resonant frequency is the maximum, the more the external frequency deviates from the resonant interval, the smaller the acoustic energy release of the system will be compared to the resonance, and thus the phenomenon of high-frequency low-energy will occur. It's

also not unique phenomenon in this research, the research [33] showed that the deeper penetration state distribute more energy in the high-frequency, which indicated a more intense energy emission in AE event.

According to the analyses above, the low-frequency high-energy and high-frequency low-energy components can be explained by the derived formulas and the physical regulation of laser welding. In turn, the monitoring data qualitatively verified the accuracy of the derived formulas. These all showed that the proposed method of “electro-mechano-acoustical” analogy theory based on keyhole pressure balance matches oscillation mechanism and monitoring data.

5. Conclusions

In this study, we coupled process monitoring methods with a novel keyhole oscillation model. The “electro-mechano-acoustical” analogy theory was used to solve keyhole bottom pressure balance equation, and resonant frequency as well as magnification amplification factor formulas of keyhole oscillation were derived. The results showed that the keyhole oscillation mechanism can match the monitoring data features and explain the observed phenomena. The conclusions are as follows:

1. The resonant frequency and the magnification amplification factor were deduced, which showed that the resonant frequency in two lateral profile situations is both related to the keyhole width and depth, and the magnification amplification factor is inversely proportional to the square of the resonant frequency;
2. The acoustic signal features and keyhole image features were extracted by the proposed feature processing methods, providing strong proofs for the validation of the derived formulas. The results showed that there are low-frequency high-energy component and high-frequency low-energy

component in time-frequency maps, corresponding with the qualitative derivation;

3. The derived formulas analyzed the phenomena in time-frequency maps of the acoustic signals. The frequency response of the resonance triggered by the keyhole oscillation is the main reason for low-frequency high-energy phenomenon. The weaker frequency response corresponding to the low-energy, and the high-frequency component was attributed to the higher frequencies obtained from the higher laser energy input.

The current research focused mainly on the qualitative analysis on the oscillation mechanism. Aiming at different materials and different welding conditions (lap or butt joints), the future study will apply these derived formulas for more laser welding application situation, which can further add more correction factor to the formulas and achieve the quantitative analysis. It will provide physics information to the data-driven prediction in traditional monitoring, and will provide novel method to deliver better theory of studying keyhole oscillation in laser welding process.

CRediT authorship contribution statement

Leshi Shu: Writing – review & editing. **Zhongyi Luo:** Writing – original draft, Investigation, Conceptualization. **Shaoning Geng:** Validation. **Ping Jiang:** Supervision, Funding acquisition, Formal analysis. **Di Wu:** Visualization, Resources, Methodology. **Deyuan Ma:** Software, Data curation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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